

CE 310 — Week 9 Assignment

Week: 9 | Topic: Simple Linear Regression — Diagnostics and Application Points: 60 pts total (3 problems)
Due: Wednesday of Week 10 by 9:00 AM — submit to D2L

Overview

The lab built a simple linear regression model predicting hourly cooling energy from outdoor temperature for a DOE Reference Medium Office building in Tucson. This assignment goes beyond the lab by examining the model's limitations, applying it to track-specific design decisions, and asking you to reflect critically on when a single linear model is — and is not — sufficient.

You will need: your completed lab notebook (Week9_Lab.ipynb), the DOE dataset (Week5_DOE_MedOffice_ZoneTemp_2023.csv) and a Python environment (Google Colab or local).

How to Submit

WARNING: Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Major declaration	Set MAJOR = "CE" or MAJOR = "ArcE" in the Setup cell — this determines which Problem 2 track is graded
Print lines	Do not edit or delete any <code>print(f"ANSWER_... = ...")</code> line
Written responses	Put each in its own markdown cell immediately after the code cell it responds to
File name	lastname_firstname_week9.ipynb e.g. smith_jane_week9.ipynb
Submission	Upload the .ipynb file to D2L Dropbox before the deadline

Setup cell (run first):

```
import pandas as pd
import numpy as np
from scipy import stats
```

MAJOR = "CE" # or "ArcE" — declares your Problem 2 track

```
df = pd.read_csv("Week5_DOE_MedOffice_ZoneTemp_2023.csv")
slope, intercept, r_value, p_value, std_err = stats.linregress(
    df['Outdoor_Temp_F'], df['Cooling_kWh']
)
print(f"MAJOR = {MAJOR}")
```

Problem 1 — Regression Diagnostics (30 pts)

Do not repeat the lab computations. Use the slope, intercept, r , R^2 , and residual plot from your completed lab notebook as starting points.

P1a (8 pts, written) — Residuals at Low Temperature

In your B8 residual plot, all residuals at outdoor temperatures below approximately 45°F are positive (the model predicts positive cooling energy, but the actual value is 0 kWh because the chiller is off).

In approximately 75 words, in a markdown cell, answer both of the following:

1. What does this systematic positive-residual pattern reveal about the validity of the linear model at low outdoor temperatures?
2. What type of regression model (not simple linear) might produce better predictions across the full temperature range? Name the approach and briefly explain why it would improve fit.

Graded on accuracy of model critique and appropriateness of proposed alternative.

P1b (8 pts) — Chiller Capacity Check

The building's chiller has a rated capacity of 200 tons (1 ton of refrigeration = 3.517 kWh/hr of cooling).

Using your regression model (slope β_1 and intercept β_0 from the lab):

(a) Predicted cooling demand at the mean outdoor temperature of 69°F

```
ANSWER_P1b_pred_69 = round(slope * 69 + intercept, 2)
print(f"ANSWER_P1b_pred_69 = {ANSWER_P1b_pred_69}")
```

(b) Remaining capacity above the 69°F baseline prediction

```
chiller_capacity = 200 * 3.517
ANSWER_P1b_remaining = round(chiller_capacity - ANSWER_P1b_pred_69, 2)
print(f"ANSWER_P1b_remaining = {ANSWER_P1b_remaining}")
```

(c) Additional degrees F needed to exhaust remaining capacity

```
ANSWER_P1b_deltaT = round(ANSWER_P1b_remaining / slope, 2)
print(f"ANSWER_P1b_deltaT = {ANSWER_P1b_deltaT}")
```

(d) Saturation temperature = 69 + deltaT

```
ANSWER_P1b_saturation_temp = round(69 + ANSWER_P1b_deltaT, 2)
print(f"ANSWER_P1b_saturation_temp = {ANSWER_P1b_saturation_temp}")
```

In a markdown cell (2–3 sentences): does this saturation temperature exceed the maximum observed outdoor temperature in the dataset ($\approx 106^\circ\text{F}$)? What does this imply about whether the 200-ton chiller can ever be overwhelmed by Tucson's climate?

P1c (7 pts) — Summer-Only Regression

Run a separate simple linear regression using only the summer rows (Season == 'Summer').

```
summer_df = df[df['Season'] == 'Summer']
ANSWER_P1c_slope, ANSWER_P1c_intercept, s_r, s_p, s_se = stats.linregress(
    summer_df['Outdoor_Temp_F'], summer_df['Cooling_kWh'])
```

```

)
ANSWER_P1c_slope = round(ANSWER_P1c_slope, 4)
ANSWER_P1c_intercept = round(ANSWER_P1c_intercept, 4)
ANSWER_P1c_r2 = round(s_r**2, 4)
print(f"ANSWER_P1c_slope = {ANSWER_P1c_slope}")
print(f"ANSWER_P1c_intercept = {ANSWER_P1c_intercept}")
print(f"ANSWER_P1c_r2 = {ANSWER_P1c_r2}")

```

In a markdown cell: Is Summer R^2 higher or lower than the all-year R^2 ? Provide a physical explanation for the direction of change. (Hint: consider what other variables vary during summer but are relatively constant in a summer-only subset.)

P1d (7 pts) — Business Hours Regression

Run a separate regression using only occupied hours (`BusinessHours == 'Yes'`).

```

biz_df = df[df['BusinessHours'] == 'Yes']
ANSWER_P1d_slope, ANSWER_P1d_intercept, b_r, b_p, b_se = stats.linregress(
    biz_df['Outdoor_Temp_F'], biz_df['Cooling_kWh']
)
ANSWER_P1d_slope = round(ANSWER_P1d_slope, 4)
ANSWER_P1d_intercept = round(ANSWER_P1d_intercept, 4)
ANSWER_P1d_r2 = round(b_r**2, 4)
print(f"ANSWER_P1d_slope = {ANSWER_P1d_slope}")
print(f"ANSWER_P1d_intercept = {ANSWER_P1d_intercept}")
print(f"ANSWER_P1d_r2 = {ANSWER_P1d_r2}")

```

In a markdown cell: Compare the occupied-hours slope to the all-year slope. Does occupancy amplify or dampen the temperature-cooling relationship? Explain physically why: when the building is occupied, what additional cooling loads are present beyond the envelope heat gain captured by outdoor temperature alone?

Problem 2 — Track Application (18 pts)

Complete only the track matching your declared MAJOR. CE → Track A. ArcE → Track B.

Track A — CE: ASHRAE 90.4 Cooling Capacity Requirement (18 pts)

ASHRAE Standard 90.4 (Energy Standard for Data Centers and Other Computing Facilities) requires that mechanical cooling capacity shall not be less than the load at the 0.4% design condition — the outdoor dry-bulb temperature exceeded by only 0.4% of annual hours (35 hours out of 8,760).

(A2-a, 6 pts) Identify the 0.4% design temperature using the 99.6th percentile, and predict the cooling demand there using your regression model:

```

ANSWER_A2a_design_temp = round(df['Outdoor_Temp_F'].quantile(0.996), 1)
ANSWER_A2a_pred_cooling = round(slope * ANSWER_A2a_design_temp + intercept, 2)
print(f"ANSWER_A2a_design_temp = {ANSWER_A2a_design_temp}")
print(f"ANSWER_A2a_pred_cooling = {ANSWER_A2a_pred_cooling}")

```

(A2-b, 6 pts) Compute the annual mean cooling demand directly from the dataset, and the safety factor between the design load and that mean:

```
ANSWER_A2b_annual_mean_cooling = round(df['Cooling_kWh'].mean(), 2)
ANSWER_A2b_safety_factor = round(ANSWER_A2a_pred_cooling / ANSWER_A2b_annual_mean_cooling, 2)
print(f"ANSWER_A2b_annual_mean_cooling = {ANSWER_A2b_annual_mean_cooling}")
print(f"ANSWER_A2b_safety_factor = {ANSWER_A2b_safety_factor}")
```

(A2-c, 6 pts) A structural engineer sized the cooling system using the annual average outdoor temperature as input to the cooling model (i.e., predicted cooling at the mean outdoor temp instead of the design temp):

```
ANSWER_A2c_avg_temp_pred = round(slope * df['Outdoor_Temp_F'].mean() + intercept, 2)
ANSWER_A2c_factor_vs_avg = round(ANSWER_A2a_pred_cooling / ANSWER_A2c_avg_temp_pred, 2)
print(f"ANSWER_A2c_avg_temp_pred = {ANSWER_A2c_avg_temp_pred}")
print(f"ANSWER_A2c_factor_vs_avg = {ANSWER_A2c_factor_vs_avg}")
```

Note: ANSWER_A2c_avg_temp_pred will come out equal to ANSWER_A2b_annual_mean_cooling — that's not a coincidence, it's a property of least-squares regression (the fitted line always passes through the point (mean x, mean y)). Worth reflecting on in your written response.

In a markdown cell (2–3 sentences): what consequence would this under-sizing have for a data center during a heat wave?

Track B — ArcE: ASHRAE 55 Adaptive Thermal Comfort (18 pts)

ASHRAE Standard 55 (Thermal Environmental Conditions for Human Occupancy) includes an adaptive thermal comfort model for naturally ventilated spaces. The comfort zone upper limit can be approximated as:

$$T_{\text{comfort, max}} = 0.31 \times T_{\text{outdoor, prevailing mean}} + 60.4 \text{ (°F)}$$

(Equation: ASHRAE 55-2023 80% acceptability upper limit, $T(^{\circ}\text{C}) = 0.31 \cdot T_{\text{pma}}(^{\circ}\text{C}) + 21.3$, converted to $^{\circ}\text{F}$: $T(^{\circ}\text{F}) = 0.31 \cdot T_{\text{pma}}(^{\circ}\text{F}) + 32 \times 0.69 + 21.3 \times 1.8 = 0.31 \cdot T_{\text{pma}}(^{\circ}\text{F}) + 60.4$)

(B2-a, 6 pts, manual — plot required) Using this equation, compute $T_{\text{comfort_max}}$ for outdoor temperatures ranging from 40°F to 110°F in 1°F increments. Plot T_{outdoor} (x-axis) vs. $T_{\text{comfort_max}}$ (y-axis). Add a horizontal reference line at 80°F (a commonly cited upper limit for comfort compliance in mixed-mode buildings).

```
T_outdoor = np.arange(40, 111, 1)
T_comfort_max = 0.31 * T_outdoor + 60.4
# your plot code here
```

(B2-b, 6 pts) Identify the outdoor temperature threshold above which $T_{\text{comfort_max}}$ exceeds 80°F. Solve algebraically, then compute:

$$0.31 \times T_{\text{outdoor}} + 60.4 > 80 \implies T_{\text{outdoor}} > ?$$

```
ANSWER_B2b_threshold = round((80 - 60.4) / 0.31, 1)
print(f"ANSWER_B2b_threshold = {ANSWER_B2b_threshold}")
```

Show the algebra in a markdown cell alongside the code.

(B2-c, 6 pts) Count how many hours in the Tucson dataset have an outdoor temperature above this threshold:

```
ANSWER_B2c_hours_above = int((df['Outdoor_Temp_F'] > ANSWER_B2b_threshold).sum())
ANSWER_B2c_frac_above = round(ANSWER_B2c_hours_above / len(df) * 100, 2)
print(f"ANSWER_B2c_hours_above = {ANSWER_B2c_hours_above}")
print(f"ANSWER_B2c_frac_above = {ANSWER_B2c_frac_above}")
```

In a markdown cell: for each of those hours, the adaptive model predicts that an 80°F indoor limit would be exceeded in a naturally ventilated space. What does this imply for the feasibility of naturally ventilated offices in Tucson's climate?

Problem 3 — Written Reflection (12 pts, written)

Write a 100–150 word reflection (no code required) in a markdown cell, addressing the following prompt:

Describe what your residual plot from B8 reveals about the validity of using a single linear regression model to predict cooling energy for this building throughout the entire year. Cite at least two specific quantitative observations from your analysis (e.g., a specific temperature range, a residual magnitude, an R^2 value, or a count of affected hours). Then propose one modification to the modeling approach that would address the largest structural limitation you identified.

Grading criteria (12 pts total): - Two quantitative observations cited (4 pts — 2 pts each) - Accurate interpretation of what the observations reveal about model validity (4 pts) - Proposed modification is specific, feasible, and addresses the identified limitation (4 pts)

Submission Checklist

- ☐ Setup cell run with MAJOR declared
- ☐ P1a: written response ~75 words
- ☐ P1b: all four ANSWER_ values printed + written discussion
- ☐ P1c: ANSWER_ slope/intercept/r2 printed + written comparison
- ☐ P1d: ANSWER_ slope/intercept/r2 printed + written comparison
- ☐ P2 (only your declared track): all ANSWER_ values printed, plot (Track B only), and written responses
- ☐ P3: reflection, 100–150 words, two quantitative observations, one proposed modification